Sre Cibi Varsshan K

C++ / GPU / Graphics / Game Programmer
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SUMMARY

Final-year Computer Science student and C++ programmer building performance-oriented software, GPU compute pipelines, and real-time rendering systems, alongside gameplay programming in C# and Unity. Hands-on experience with GPU compute, real-time rendering, shader programming, physics-based movement, modular game systems, parallel computation, and graphics programming. Seeking C++/GPU/Graphics Programmer or Unity Game Programmer roles in performance engineering, rendering, or gameplay systems.

EDUCATION

VIT Chennai Expected 2027
B.Tech in Computer Science
Relevant coursework: Computer Graphics, Data Structures & Algorithms, Machine Learning, Computer Vision

PROJECTS

GPU-Aided Bulk Image Encryption (ChaCha20) *C++, OpenGL, GLSL*

- Implemented ChaCha20 image encryption as an OpenGL compute pipeline, mapping ARX-based keystream generation across GPU workgroups.
- Designed Shader Storage Buffer Object (SSBO)-based buffer handling for batch image processing and metadata-driven reconstruction for lossless decryption.
- Applied parallel computation and GPU buffer management to move encryption work from CPU-side iteration to GPU execution.

GPU Ray Marcher *C++, OpenGL, GLSL*

- Built a real-time GPU ray marcher using compute and fragment-shader pipelines for procedural 3D rendering.
- Implemented signed distance functions, lighting, and shader-based scene evaluation with rendering work executed on the GPU.

CPU Ray Tracer *C++*

- Built a CPU ray tracer from scratch implementing ray-object intersection, materials, reflections, and light-transport effects.
- Applied rendering mathematics and recursive ray evaluation to model physically based rendering concepts.

Deep Q-Network Space Shooter *Python, PyTorch*

- Built a DQN agent for a 2D space-shooter environment using a convolutional network to learn action values directly from visual state inputs.
- Implemented experience replay, target-network stabilization, environment simulation, and reward shaping for autonomous gameplay decisions.

Doofus Diary *Unity 6, C#*

- Built a 3D platform-survival game with modular gameplay systems, dynamic platforms, progressive difficulty, scoring, UI, audio, and player progression.
- Implemented single- and double-jump progression, Rigidbody-based movement, and a spring-damper camera with momentum-aware following.
- Structured player movement, platform spawning, game state, scoring, UI, audio, and JSON-based configuration as focused gameplay systems; shipped a playable WebGL build.

Physics-Based FPS Controller & Rendering Sandbox *Unity, C#*

- Built a physics-oriented FPS controller with custom acceleration, friction, air control, jumping, and momentum-based traversal.
- Implemented Quake-style movement features including coyote time and jump buffering, with runtime physics and responsive player movement as primary design constraints.
- Used the project as a real-time rendering sandbox for shader-driven visual effects and lighting.

Slot Machine *Unity 6, C#, ShaderLab, HLSL*

- Built a 3-reel slot-machine game with weighted random number generation, staggered reel animation, pixel-perfect alignment, and data-driven symbol configuration.
- Developed a custom URP post-processing shader combining palette quantization, ordered dithering, and Sobel edge detection.
- Shipped a playable WebGL build combining gameplay logic, animation, UI, and custom GPU shader effects.

TECHNICAL SKILLS

Programming Languages: C++, C, C#, Python, Bash
GPU & Graphics Programming: OpenGL, GLSL, Compute Shaders, Shader Storage Buffer Objects (SSBO), HLSL, ShaderLab, Ray Tracing, Ray Marching, Real-Time Rendering
Game Development: Unity 6, Gameplay Systems, Rigidbody Physics, Physics-Based Movement, Game AI, UI Systems, Data-Driven Configuration, WebGL
Algorithms & Systems: Data Structures & Algorithms, Parallel Computing, Memory Management, Buffer Management
Tools & Platforms: Linux, Git, Version Control